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Examination paper for TDT4145 Data Modelling, Databases and Database Management Systems

Date: 8. May 2026

Time: 09:00-12:00 (09am-12am)

Course contact: Svein Erik Bratsberg, Theodoros Chondrogiannis

Present at the exam location: NO.

Permitted examination support material: D: No printed or hand-written support material is allowed. A specific basic calculator is allowed.

OTHER INFORMATION

Read the questions carefully. There is a comment problem at the end where you may state assumptions.

If there are direct errors or omissions in the assignment set and you cannot make your own assumptions, please refer to the information about complaints regarding formal errors on the NTNU website "Explanation of grades and appeals".

SPECIFIC INFORMATION FOR YOUR COURSE

No paper drawings

This exam does not include hand drawings. If you receive hand drawing sheets, this is by mistake. **You will not be able to submit the sheets, and they will not be graded.**

Weighting:

The weighting is given in the heading of each problem.

Change the language of the question set:

If the question set is given in both English and Norwegian, you can change the language by going to the "hamburger menu" in the top hand-right corner and click "Language".

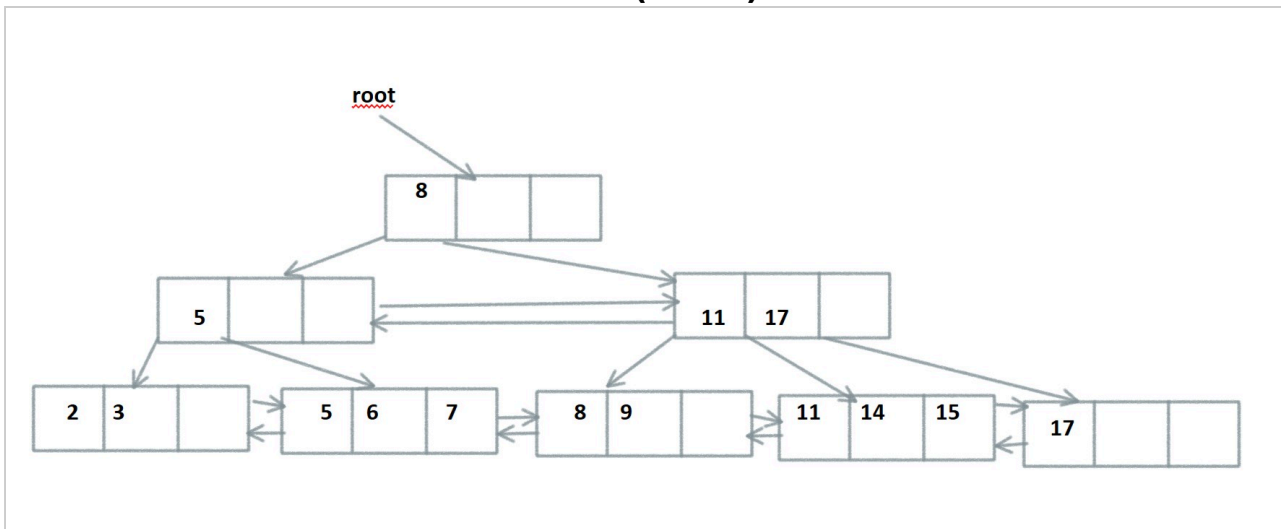
Withdrawing from the exam:

If you wish to submit a blank test/withdraw from the exam for another reason, go to the "hamburger menu" in the top right-hand corner and click "Submit blank". This cannot be undone, even if the test is still open.

Access to your answers:

After the exam, you can find your answers under previous tests in Inspera.

1 Problem 1: B+-tree insertions (10 %)



Which sequence of inserts gives the B+-tree above?

Select one alternative:

- ☒ 2, 5, 3, 11, 7, 8, 6, 9, 17, 14, 15
- ☐ No other is correct
- ☐ 2, 3, 5, 6, 7, 8, 9, 11, 14, 15, 17
- ☐ 6, 7, 8, 2, 3, 5, 17, 15, 14, 11, 9

Maximum marks: 10

2 Problem 2: B+-tree theory 1 (5 %)

Why does the split of B+-tree blocks happen firstly without considering where the new key to be inserted ends, and then you can insert the new key?

Select one or more alternatives:

- ☒ The split should both work for inserts of new data and for extensions of existing data (i.e. updates of VARCHAR/TEXT). Thus, no new keys exists in the latter case.
- ☒ During recovery we want to let the split be redone before the record operation is redone and inserted.
- ☐ This creates an optimal allocation of the two blocks such that they have always equal number of the records in the two blocks after the split.
- ☐ This lets the blocks be completely filled, making the fill degree close to 100 %.
- ☐ No other alternative is correct.

Maximum marks: 5

3 Problem 3: B+-tree theory 2 (5 %)

Why is the fill degree of B+-tree blocks less than 100 %, typically 2/3?

Select one or more alternatives:

- ☒ Because of random key insert, it is impossible to fill blocks without using compaction.
- ☐ It is much less work to write blocks to disk when they are not filled.
- ☐ It is to save the disk for storing a lot of data each time a block is written to disk.
- ☐ Because we leave place for keys to be inserted where we expect them to end.
- ☐ No other alternative is correct.

Maximum marks: 5

4 Problem 4: Transaction theory (6 %)

What conditions define a conflict between two operations of two transactions?

Select one or more alternatives:

- ☐ They do a dirty read
- ☐ It allows undo recovery by before images.
- ☒ At least one of the operations must be a write.
- ☐ No other alternative is correct
- ☒ The operations belong to different transactions
- ☐ There are no cascading aborts (ACA)
- ☒ The operations are to the same data element

Maximum marks: 6

5 Problem 5: Access methods 1 (5 %)

We have the following table:

Student (pno, studno, name, address).

Each record is 80 bytes. We have 10 060 students in the table. Each data block is 4096 bytes.

The table is stored in a clustered B+-tree using the primary key **pno** as search key.

Assume studno to be unique as well. Assume the B+-tree's blocks to have a fill degree of 2/3.

Assume pno to be 4 bytes, a BlockId to be 8 bytes and a RecordId to be 12 bytes,

There are two queries issued towards the database:

(1) SELECT * FROM Student WHERE pno = <person number>;

(2) SELECT * FROM Student WHERE studno = <student number>;

60 % of the queries are of type (1) and 40 % of the queries are of type 2.

How many blocks are accessed in average per query using the storage structure just mentioned?

Select one alternative:

☐ 6

☒ 61.8

☐ 301

☐ 298

☐ No other alternative is correct

Maximum marks: 5

6 Problem 6: Access methods 2 (5 %)

We have the following table:

Student (pno, studno, name, address).

Each record is 80 bytes. We have 10 060 students in the table. Each data block is 4096 bytes.

The table is stored in a heap file and using an unclustered B+-tree to index the heap file using the primary key **pno** as search key. Assume studno to be unique as well. Assume the B+-tree's blocks to have a fill degree of 2/3.

Assume pno to be 4 bytes, a BlockId to be 8 bytes and a RecordId to be 12 bytes,

There are two queries issued towards the database:

(1) SELECT * FROM Student WHERE pno = <person number>;

(2) SELECT * FROM Student WHERE studno = <student number>;

60 % of the queries are of type (1) and 40 % of the queries are of type 2.

How many blocks are accessed in average per query using the storage structure just mentioned?

Select one alternative:

- ☐ 106
- ☐ No other alternative is correct
- ☐ 107
- ☐ 44
- ☒ 41.4

Maximum marks: 5

7 Problem 7: Access methods 3 (5 %)

We have the following table:

Student (pno, studno, name, address).

Each record is 80 bytes. We have 10 060 students in the table. Each data block is 4096 bytes.

The table is stored in a clustered B+-tree using the primary key **studno** as search key.

Assume pno to be unique as well. Assume the B+-tree's blocks to have a fill degree of 2/3.

Assume studno to be 4 bytes, a BlockId to be 8 bytes and a RecordId to be 12 bytes,

There are two queries issued towards the database:

(1) SELECT * FROM Student WHERE pno = <person number>;

(2) SELECT * FROM Student WHERE studno = <student number>;

60 % of the queries are of type (1) and 40 % of the queries are of type 2.

How many blocks are accessed in average per query using the storage structure just mentioned?

Select one alternative:

- ☐ 6
- ☐ No other alternative is correct
- ☐ 298
- ☒ 91.2
- ☐ 61.8

Maximum marks: 5

8 Problem 8: Access methods 4 (5 %)

We have the following table:

Student (pno, studno, name, address).

Each record is 80 bytes. We have 10 060 students in the table. Each data block is 4096 bytes.

The table is stored in a clustered hash file using the primary key **pno** as search key.

Assume studno to be unique as well.

Assume pno to be 4 bytes, a BlockId to be 8 bytes and a RecordId to be 12 bytes. Assume a clustered hash file's blocks to have a fill degree 0.8, and that the average lookup cost on search key to be 1.2.

There are two queries issued towards the database:

(1) SELECT * FROM Student WHERE pno = <person number>;

(2) SELECT * FROM Student WHERE studno = <student number>;

60 % of the queries are of type (1) and 40 % of the queries are of type 2.

How many blocks are accessed in average per query using the storage structure just mentioned?

Select one alternative:

☒ 51.12

☐ 1.2

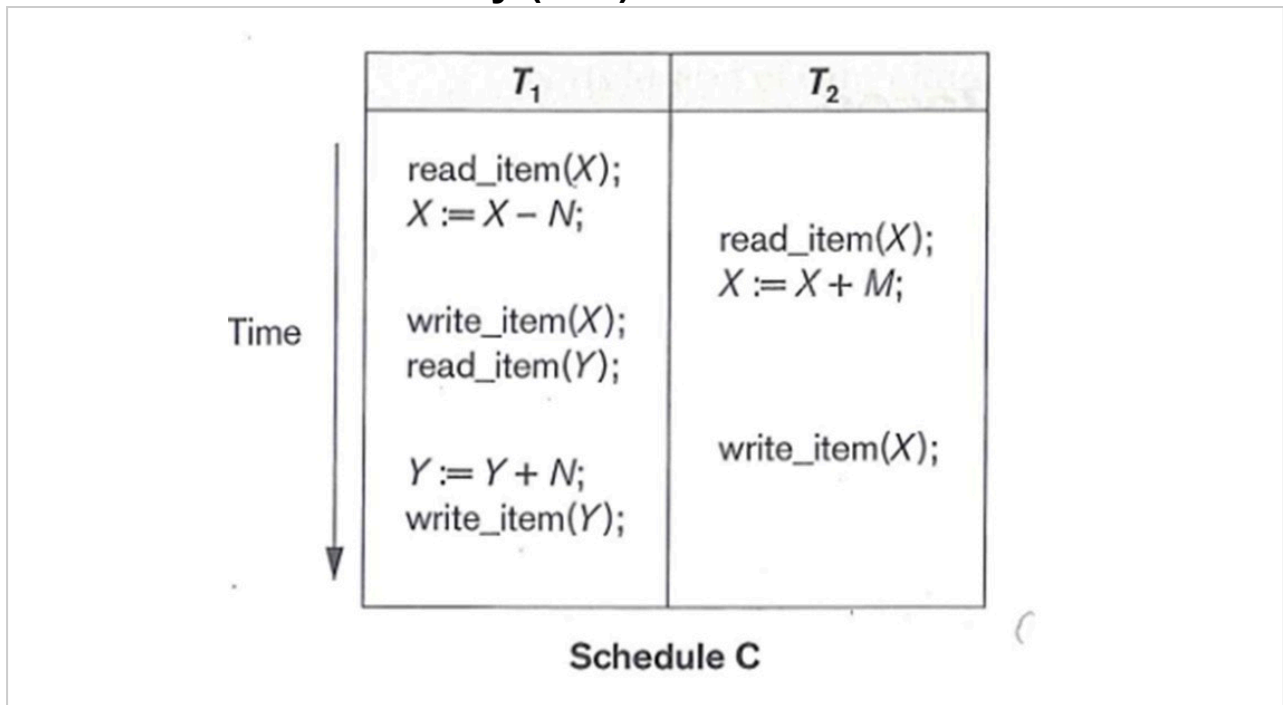
☐ 253.2

☐ 2.4

☐ No other alternative is correct,

Maximum marks: 5

9 Problem 9: Serializability (5 %)



The figure shows a non-serializable schedule of operations. `read_item` and `write_item` operate directly on disk. Assume $X=90$, $Y=90$, $N=3$ and $M=2$ at start of the execution. What are the values of X and Y after the non-serializable execution?

Select one alternative:

- ☐ No other alternative is correct
- ☐ $X=92$, $Y=91$
- ☐ $X=89$, $Y=93$
- ☒ $X=92$, $Y=93$
- ☐ $X=90$, $Y=93$

Maximum marks: 5

10 Problem 10: 2PL and lock setting (5 %)

We have the following sequence of operations which are scheduled for execution using 2PL and the rigorous model for locking. In what sequence does the transactions commit: r1(A); r2(B); w1(B); w1(C); r3(C); r2(A); c1; c2; c3

Select one alternative:

- ☐ We get a deadlock between T1 and T2.
- ☐ No other alternative is correct
- ☐ T1; T3; T2
- ☒ T2; T3; T1
- ☐ T1; T2; T3
- ☐ T2; T1; T3
- ☐ T3; T1; T2
- ☐ T3; T2; T1

Maximum marks: 5

11 Problem 11: Recovery (4 %)

LSN	PrevLSN	TransactionID	Operation	PageID
101	99	T1	Update	B
102	0	T2	Update	B
103	101	T1	Update	A
104	102	T2	Update	A
105	103	T1	Update	C
106			Ckpt start	
107			Ckpt end	
108	105	T1	Commit	
109	104	T2	Update	D
110	0	T3	Update	E
111	109	T2	Commit	

The following log was found after a crash. We use ARIES recovery. In the checkpoint log record with LSN=107 the following transaction table was found (T1, LastLSN=105, state=Active), (T2, LastLSN=104, state=Active) and the following DPT (Dirty Page Table) was found: (PageID=B, recLSN=102), (PageID=C, recLSN=105). In addition, we know that when running the log entry with LSN=105, the space for PageID=A was "stolen" from the buffer. Based on this and the log above, which data blocks (Pages) can we say with certainty were written out before the crash? Note that the page may have become dirty after the write.

Select one or more alternatives:

☒ B

☐ No other alternative is correct

☐ D

☐ C

☒ A

☐ E

Maximum marks: 4

12 Problem 12: Relational Model (4%)

Choose the sentence that is **wrong** or the option "All sentences are correct" if there are no wrong statements.

Select one alternative:

- ☐ All sentences are correct.
- ☐ The special value NULL is a member of every domain.
- ☒ The join operator can be defined using using projection and cartesian product operators.
- ☐ A superkey K is a candidate key if K is minimal.
- ☐ A database instance is a snapshot of the data in a database at a given point in time.

Maximum marks: 4

13 Problem 13: Relational Algebra (4%)

Assume that R is a relation, att is some attributes of R , and $cond$ is a condition on R . When will the following be true?

$$\sigma_{cond}(\pi_{att}(R)) = \pi_{att}(\sigma_{cond}(R))$$

Select one alternative:

- ☒ It will be true only when the conditions in $cond$ refer to just the attributes in att .
- ☐ It will always be true.
- ☐ None of the other alternatives is correct.
- ☐ It will be true only when R is a binary relation, with two attributes.
- ☐ It will be true only when there are two attributes in att .

Maximum marks: 4

14 Problem 14: SQL I (6%)

You are given the following schema and tables of a relational database (*sid* in *room* is a foreign key referencing *sid* in *student*):

student(*sid*, *sname*, *semail*, *syear*)

room(*rid*, *rrent*, *rarea*, *sid*)

student

sid	sname	semail	syear
1	Carie	carie@em.com	2003
2	Terrye	terrye@em.com	2002
3	Camella	camella@em.com	2005
4	Andie	andie@em.com	2002
5	Vinny	vinny@em.com	2000

room

rid	rrent	rarea	sid
1	10000	43	1
2	11500	37	NULL
3	10300	31	NULL
4	11800	44	2
5	10500	43	NULL
6	12400	36	3
7	13800	39	4
8	10700	40	5

What is the result of the following query?

```

SELECT rrent
FROM student NATURAL JOIN room
WHERE syear > ANY (
  SELECT syear
  FROM student
  WHERE sname LIKE 'C%'
)
ORDER BY syear DESC
LIMIT 1

```

Fill in your answer here:

12400

Maximum marks: 6

15 Problem 15: SQL II (6%)

You are given the following schema and tables of a relational database (*sid* in *room* is a foreign key referencing *sid* in *student*):

student(*sid*, *sname*, *semail*, *syear*)

room(*rid*, *rrent*, *rarea*, *sid*)

student

sid	sname	semail	syear
1	Carie	carie@em.com	2003
2	Terrye	terrye@em.com	2002
3	Camella	camella@em.com	2005
4	Andie	andie@em.com	2002
5	Vinny	vinny@em.com	2000

room

rid	rrent	rarea	sid
1	10000	43	1
2	11500	37	NULL
3	10300	31	NULL
4	11800	44	2
5	10500	43	NULL
6	12400	36	3
7	13800	39	4
8	10700	40	5

What is the result of the following query?

```

SELECT b.sid
FROM room AS a, room AS b
WHERE a.rid <> b.rid
AND a.rarea = b.rarea
AND b.sid IS NOT NULL
ORDER BY b.rarea
LIMIT 1

```

Fill in your answer here:

Maximum marks: 6

16 Problem 16: SQL III (6%)

You are given the following schema and tables of a relational database (*sid* in *room* is a foreign key referencing *sid* in *student*):

student(*sid*, *sname*, *semail*, *syear*)

room(*rid*, *rrent*, *rarea*, *sid*)

student

sid	sname	semail	syear
1	Carie	carie@em.com	2003
2	Terrye	terrye@em.com	2002
3	Camella	camella@em.com	2005
4	Andie	andie@em.com	2002
5	Vinny	vinny@em.com	2000

room

rid	rrent	rarea	sid
1	10000	43	1
2	11500	37	NULL
3	10300	31	NULL
4	11800	44	2
5	10500	43	NULL
6	12400	36	3
7	13800	39	4
8	10700	40	5

What is the result of the following query?

```
SELECT count(*)
FROM (
  SELECT rid AS aid, rarea
  FROM room
) AS a NATURAL JOIN (
  SELECT rid AS bid, rarea
  FROM room
) AS b
WHERE aid <> bid
```

Fill in your answer here:

2

Maximum marks: 6

17 Problem 17: SQL IV (6%)

You are given the following schema and tables of a relational database (*sid* in *room* is a foreign key referencing *sid* in *student*):

student(*sid*, *sname*, *semail*, *syear*)

room(*rid*, *rrent*, *rarea*, *sid*)

student

sid	sname	semail	syear
1	Carie	carie@em.com	2003
2	Terrye	terrye@em.com	2002
3	Camella	camella@em.com	2005
4	Andie	andie@em.com	2002
5	Vinny	vinny@em.com	2000

room

rid	rrent	rarea	sid
1	10000	43	1
2	11500	37	NULL
3	10300	31	NULL
4	11800	44	2
5	10500	43	NULL
6	12400	36	3
7	13800	39	4
8	10700	40	5

What is the result of the following query?

```

SELECT count(sname)
FROM student NATURAL JOIN room
WHERE rrent > (
  SELECT max(rrent)
  FROM room
  WHERE rarea >= 40
  AND sid IS NULL
)
GROUP BY syear
HAVING syear > 2000
ORDER BY syear
LIMIT 1

```

Fill in your answer here:

2

Maximum marks: 6

18 Problem 18: ER Constraints (4%)

Which of the following concepts is **NOT** modeled explicitly in the standard notation of ER diagrams?

Select one alternative:

- ☒ Foreign keys.
- ☐ Primary keys.
- ☐ All other options are modeled explicitly in an ER diagram.
- ☐ Cardinality constraints.
- ☐ Participation constraints.

Maximum marks: 4

19 Problem 19: Normalization (4%)

Atomicity of domains of attributes is related to

Select one alternative:

- ☒ 1NF
- ☐ 3NF
- ☐ BCNF
- ☐ None of the other options.
- ☐ 2NF

Maximum marks: 4

20 Problem 20: Comments (0 %)

This "task" is an opportunity to inform about circumstances, which are **absolutely necessary** to communicate to the censors, in order for them to make a correct judgement of your answers. This can for instance be assumptions that was completely unavoidable.

This is *not* for *general comments* about the exam, which can be stated on Ed or in email to the teachers.

Write your answer in the box below. Changes are saved automatically.

Format
|
B
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 $\frac{1}{2}$
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Words: 0

Maximum marks: 0